



Menstrual Health Data

TBA

Project Proposal

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	Type:	Project Proposal
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Table of Contents

Project Overview	2
Problem Statement	2
Objectives	2
Main Objective	2
Specific Objectives	2
Dataset Description	3
Demographic Information	3
Menstrual Health Data	3
Lifestyle Factors	3
Physiological Measurements	4
Analytics and Predictions	4
Analytical Challenge	4
Dashboard Development	5
Predictive Analytics	5
Optional System Development	6
Deliverables	6
Deliverable 1 – Project Plan and Initial Analysis	6
Deliverable 2 – Dashboard Prototype	7
Deliverable 3 – Predictive Analysis	7
Deliverable 4 – Final Presentation and submission	7
Reading Materials	8

	Type:	Project Proposal
	Version:	2.0

Project Overview

The goal of this project is to analyse menstrual health data and identify meaningful relationships between menstrual cycles, symptoms, and lifestyle factors.

Students will work with a dataset containing menstrual cycle records, lifestyle information, and optional wearable-device data. The project focuses on data preparation, visualization, dashboard development, and predictive analytics.

The dataset was designed to represent a heterogeneous population with different menstrual health profiles and behaviours. Students are encouraged to investigate whether distinct patterns or user groups emerge from the data and to assess which factors contribute most strongly to the observed differences.

Students are expected to investigate the data, formulate analytical questions, develop insights, and communicate their findings through reports and presentations.

The project is designed to be completed within two weeks.

Problem Statement

Menstrual health is influenced by multiple factors such as sleep quality, stress levels, physical activity, hydration, and physiological conditions.

While many health applications collect this information, it is often difficult to understand how these variables interact and affect menstrual cycle characteristics.

The challenge of this project is to explore these relationships using data analytics techniques and develop visual and predictive tools that support menstrual health understanding.

Objectives

Main Objective

Develop a data analytics solution that explores menstrual health patterns and generates actionable insights through visualization and predictive modelling.

Specific Objectives

Students should aim to:

- Understand and prepare the dataset.

	Type:	Project Proposal
	Version:	2.0

- Identify patterns and trends in menstrual cycle data.
- Analyse relationships between lifestyle factors and cycle characteristics.
- Create informative dashboards and visualizations.
- Develop predictive or analytical models.
- Communicate findings and recommendations effectively.

Dataset Description

A dataset containing approximately:

- 50 users
- 6 months of historical data per user

will be provided for the project.

The dataset combines information from multiple domains that may influence menstrual health and cycle characteristics.

Demographic Information

Basic user characteristics may include:

- Age
- Height and weight
- Medical background
- Medication usage (where available)
- Other relevant demographic information

Menstrual Health Data

Historical menstrual records may include:

- Period start and end dates
- Cycle duration
- Period duration
- Fertility-related information
- Reported symptoms
- Pain levels
- Mood indicators

Lifestyle Factors

Lifestyle information may include:

	Type:	Project Proposal
	Version:	2.0

- Sleep patterns
- Stress levels
- Physical activity
- Hydration habits
- Nutritional indicators
- Other daily wellness measurements

The dataset contains both regular and irregular cycle patterns, varying lifestyle behaviours, symptom records, and medication-related information. Relationships between these variables may exist and should be explored through analysis.

Physiological Measurements

Where available, physiological and wearable-device data may include:

- Heart rate measurements
- Resting heart rate
- Body temperature
- Activity metrics
- Fitness tracker information
- Other health-related indicators

Students are encouraged to explore the dataset, identify relevant variables, and engineer additional features that may improve analysis or predictive performance.

The exact analytical approach, feature selection, and data preparation strategy are left to the students.

Analytics and Predictions

Analytical Challenge

The dataset was designed to contain multiple user behaviours and health patterns. Students are encouraged to investigate whether distinct user groups emerge from the data and to evaluate which factors appear to influence menstrual health outcomes.

Students should perform:

- Data cleaning and preprocessing.
- Missing value analysis.
- Descriptive statistics.
- Trend analysis.

	Type:	Project Proposal
	Version:	2.0

- Outlier detection.
- Correlation analysis.

Possible research questions include:

- Are there identifiable user profiles based on cycle regularity and symptoms?
- Which lifestyle factors are most strongly associated with cycle variability?
- Can groups of users be discovered through clustering or segmentation techniques?
- Do some factors explain only part of the observed variability?
- Are there users whose behaviour does not follow the overall trends?

Students are encouraged to formulate their own hypotheses and validate them using data-driven methods.

Dashboard Development

Students must create an interactive dashboard that presents key findings from the dataset.

The dashboard should include:

- Menstrual cycle statistics.
- Lifestyle indicators.
- Symptom trends.
- User-level analysis.
- Population-level summaries.
- Predictive outputs (if applicable).

The design and content of the dashboard are left to the students. Students may use tools such as:

- Power BI
- Tableau
- Python Dash
- Streamlit
- Other suitable visualization platforms

Predictive Analytics

Students must implement at least two analytical or predictive feature.

	Type:	Project Proposal
	Version:	2.0

Possible examples include:

Option A – Next Period Prediction

Estimate the next period start date using historical cycle data.

Option B – Cycle regularity/Irregularity Detection

Identify users with unusual cycle behaviour.

Option C – Symptom Prediction

Predict symptom occurrence or severity.

Option D – Lifestyle Impact Modelling

Determine which lifestyle variables have the strongest influence on cycle characteristics.

Option E – Student-Defined Analytical model

Students may propose and justify their own analytical objective.

The methodology and tools used are left to the students.

Optional System Development

Students may optionally develop a simple application to present their dashboard and analysis results.

Examples include:

- Web application
- Mobile application
- Dashboard portal

This component is not mandatory and should only be attempted after the analytics requirements have been completed.

Deliverables

The final system should provide the following information.

Deliverable 1 – Project Plan and Initial Analysis

- Project plan

	Type:	Project Proposal
	Version:	2.0

- Dataset exploration
- Initial observations and literature analysis
- Proposed analytical objectives

Deliverable 2 – Dashboard Prototype

- Data cleaning documentation
- Exploratory analysis results
- Dashboard prototype
- Description of selected analytical approach

Deliverable 3 – Predictive Analysis

- Predictive model(s)
- Segmentation or clustering analysis
- Interpretation of discovered patterns
- Discussion of limitations

Deliverable 4 – Final Presentation and submission

10–15 minute presentation including:

- Problem definition
- Dataset overview
- Dashboard demonstration
- Key insights
- Predictive results
- Conclusions and recommendations

Submit a ZIP file containing:

- Source code and notebooks
- Dashboard files
- Dataset (if modified)
- Supporting documentation
- Instructions for execution

Include a final report (5–8 pages) describing:

- Methodology
- Data preparation process
- Visualizations
- Analytical findings

	Type:	Project Proposal
	Version:	2.0

- Predictive models
- Conclusions

Reading Materials

Here are some materials to be read before starting the project:

1. How the menstrual cycle works
2. Important health indicators
3. Lifestyle factors affecting cycles
4. Ethical considerations

[NHS inform - Period](#)

[Mayo Clinic - Menstrual Cycle](#)

[The Menstrual Cycle - a vital sign across the lifespan](#)

[The role of lifestyle on the menstrual cycle](#)



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